Lecture 1 Recap

Smart Parking System

A smart parking system manages parking spaces in a shopping mall. When a vehicle enters, the system assigns an available parking slot. When the vehicle leaves, the system calculates the parking fee based on the duration. The system supports Cars, Bikes and Trucks.

Part A — Identify Classes

Find **at least 7 classes**.

Part B — Attributes

For each class:

Identify at least **3 attributes**.

Part C — Methods

For each class:

Identify at least **2 methods**.

Part D — Inheritance

Create the inheritance hierarchy.

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Part E — Encapsulation

Choose two classes.

Identify:

- What data should be private?
- Why should it be hidden?
- Which methods should control access?

Part F — Polymorphism

Identify:

One method that should behave differently for Car, Bike and Truck.

Part G — Cohesion

Choose one class and explain:

Why does this class have high cohesion?

Part H — Coupling

Identify one unnecessary dependency between classes and suggest how to reduce it.